



MOKO Remote Gateway Configuration Manual

Version V1.4

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1. About this document

MOKO developed the mobile APP “MKScannerPro” for gateway configuration. As MOKO has many gateway types and models, the APP includes some components for different gateway types. The remote gateway means the gateway works as a scanner and remote controller for BLE devices. The device types and gateway models are as below:

APP name	APP device type	Gateway device model
MKScannerPro-> Remote Gateway	Remote gateway -32	MK107-32 MKGW-mini 01-32 MK110 Plus 01-32
	MK107Pro-32D	Only MK107Pro-32D
	MK107D Pro-35D	Only MK107D Pro-35D
	MKGW-mini 02-32D	Only MKGW-mini 02-32D
	MKGW-mini 03-20D	Only MKGW-mini 03-20D

2. LED status

Function	Action	LED Patterns
Bluetooth status (Pairing mode)	Bluetooth is advertising	Flash Blue
	Bluetooth is connected	Solid Blue
WIFI status	Connecting to the WIFI	Slowly flash Green (every 2s)
	Connecting to the server	Quickly flash Green (every 500ms)
	Connected to the server successfully	Solid Green
Downlink communication	Connected with Beacon	Solid Purple
Restore to factory settings	Press button for 10 seconds in 1 minute after powered, the gateway will be reset and enter pairing mode.	Alternately flash Blue and Green once
OTA status	OTA process	Flash Yellow
	OTA succeed	Solid Yellow
	OTA failed	Solid Red

3. Configure gateway with mobile APP

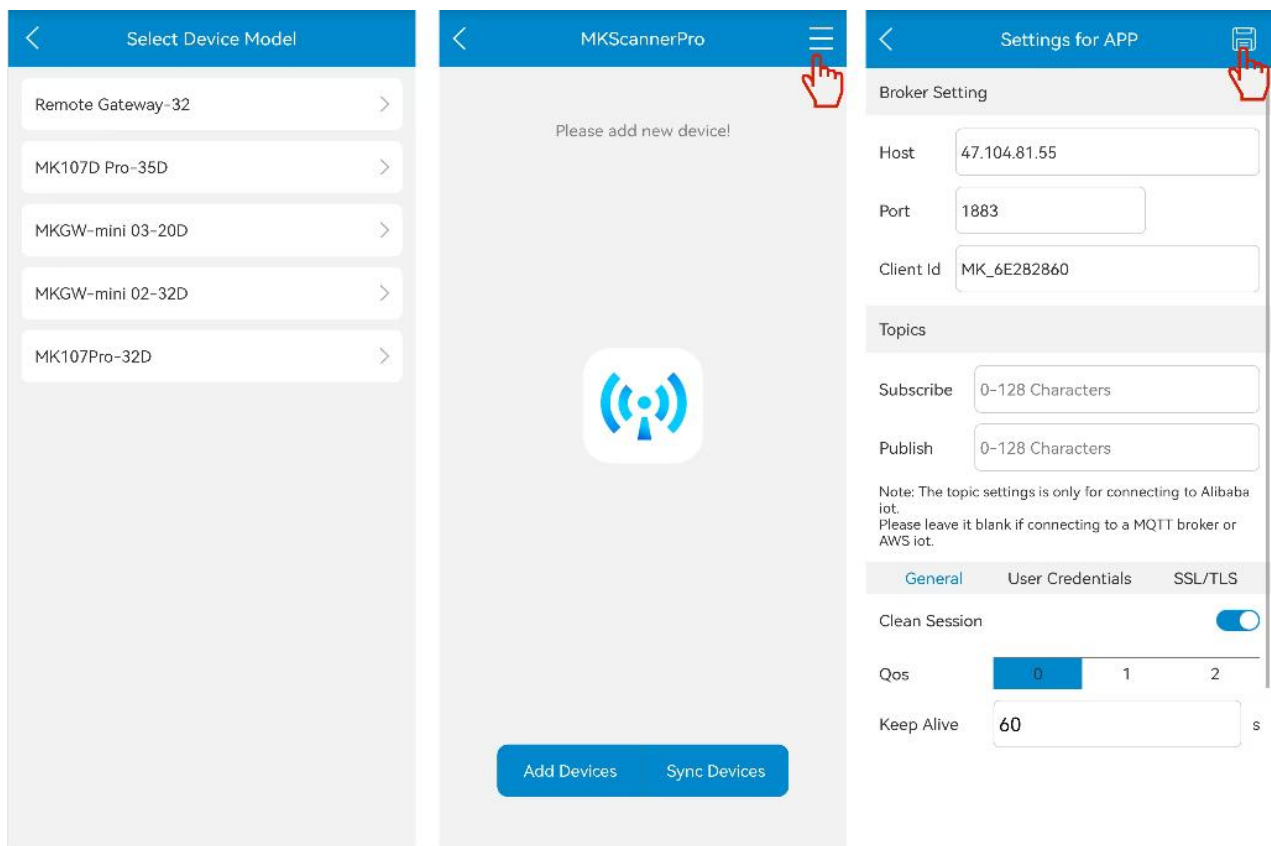
Please search and download “MKScannerPro” APP from Apple APP Store or Google play. When installing it, please allow the location and storage permission.

In the configuration, we will firstly configure the APP connecting to MQTT broker and then configure gateway connecting to MQTT broker. After the APP and gateway are both connected with MQTT broker, users can use the APP to remotely manage the gateway.

3.1 Configure mobile APP

Run the mobile APP, select “**Remote gateway**”, and select the correct device model, then we can start the configuration. If you are not sure for the device model, please find the label on the gateway to get it.

Click  to configure MQTT settings for the APP. The APP has default MQTT settings, if use the default settings for testing purpose, just click . It also allows to change the settings and configure customer server, after enter the settings and save it, app will connect to the MQTT server. If connect successfully, it will show “success”, otherwise it will show “connect failed”. If connect failed, users need to check the settings and connect the APP again.



There are three buttons in the very bottom of this page, to help users quickly complete the configuration:

- Clear all configurations: Delete all the current settings, so that users can input new settings.
- Export config file: Export the current settings from the APP, it will create a excel file and can be sent by email.
- Import config file: With the exported file, users can change the settings and import the new file to the APP, then the APP will use the new settings.

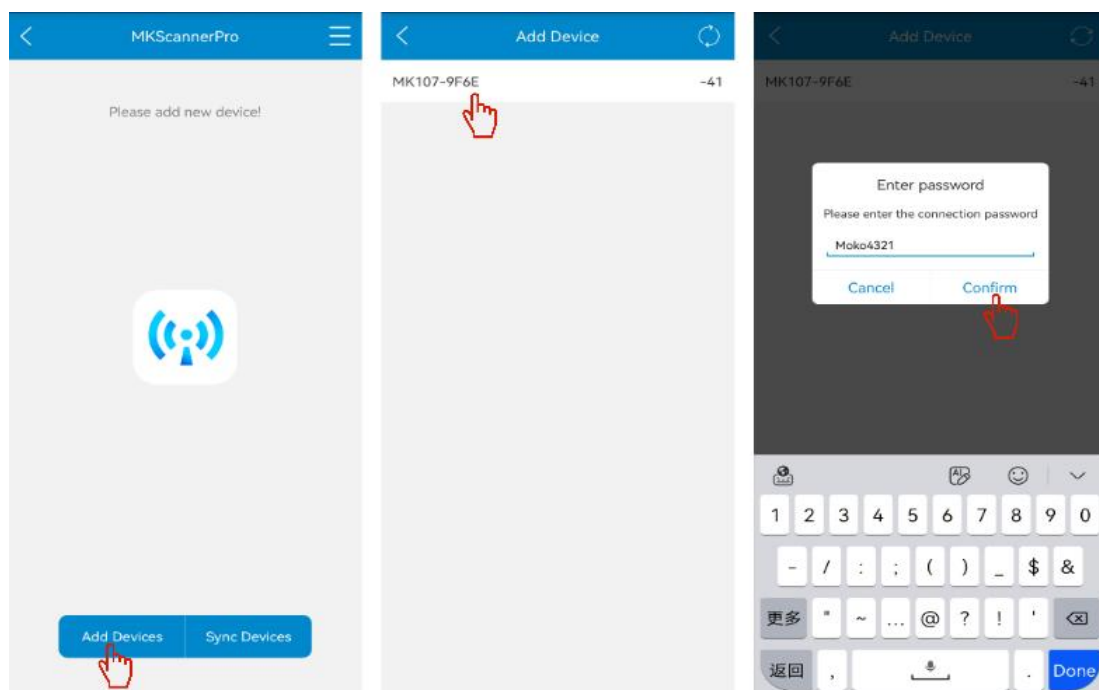
If configure with custom server, please follow these settings description as below:

Type	Parameter	Description
Broker setting	Host	The IP address or domain name of the MQTT endpoint
	Port	MQTT port
	Client id	Each device connected to the server should have a unique client id.
Topics	Subscribe	These two settings are used for connecting to Alibaba cloud. If connect to a MQTT broker or AWS IoT, please leave it blank.
	Publish	
General	Clean session	Default: Enable, range: Enable/Disable
	Qos	Quality of service. Default: 1, range: 0-2
	Keep Alive	Default: 60, range: 10-120
User Credentials	Username	If access to your server doesn't require a username and password, it can be blank.
	Password	
SSL/TLS	SSL/TLS	Enable: SSL mode. Disable: TCP mode
	Certificates	It supports three types of certificates: ➤ CA signed server certificate: no need to upload certificate ➤ CA certificate file: needs to upload CA Root certificate ➤ Self signed certificates: needs to upload CA Root certificate, client key and client certificate

If connect to AWS IoT, please see the [Appendix B: Connect to AWS IoT](#) to get more details.

3.2 Configure gateway

When the gateway LED flashes **Blue**, click “Add Devices”, we can see some device advertisements appearing on the screen. Select the correct device and enter password **Moko4321**, then the APP will connect with the gateway Bluetooth, gateway LED will turn to solid **Blue**.



The gateway advertisement name is their model name plus the last 2 bytes of mac address.

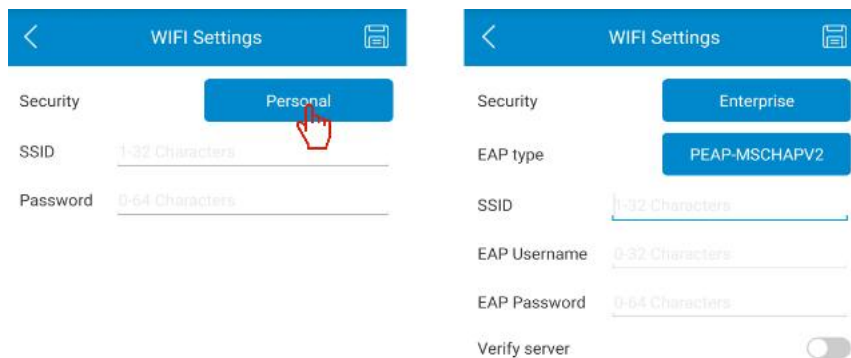
Gateway model	Gateway advertisement name
MK107-32	MK107-XXXX
MKGW-mini 01-32	MINI 01-XXXX
MK110 Plus 01-32	MK110 Plus 01-XXXX
MK107Pro-32D	MK107Pro-XXXX
MK107D Pro-35D	MK107D Pro-XXXX
MKGW-mini 02-32D	MINI 02-XXXX
MKGW-mini 03-20D	MINI 03-XXXX

2. Then we can configure the WIFI, MQTT and some other settings for the gateway. After all settings are finished, click “Connect” button, the gateway will connect to the WIFI and server, the indicator will flash **Green**. If the gateway connects to server successfully, the LED will turn to solid **Green**. If failed, the gateway will go back to pairing mode, you can configure it again.

Please note: the WIFI and MQTT settings are must-required, while other settings are optional.

3.2.1 WIFI settings

To configure WIFI information for the gateway, it supports personal and enterprise security. The personal WiFi requires only the SSID and password information. The enterprise WIFI supports different EAP types and will require different authentication information. Please see the [Appendix A: WPA2 Enterprise Security](#) to get more details.



Some differences of gateway models:

1. MK107DPro-35D and MKGW-mini 03-20D gateway support both 2.4GHZ and 5GHZ WIFI, the WIFI 5GHZ band are different in countries, therefore, it requires to set the 5GHZ wifi band according to the region. Available only on MK107DPro-35D and MKGW-mini 03-20D gateways:



2. MKGW-mini 03-20D gateway does not support enterprise security, only personal security is available.

3.2.2 MQTT settings

To configure MQTT information for the gateway. The gateway has default MQTT settings, if use default settings for testing purpose, just click .

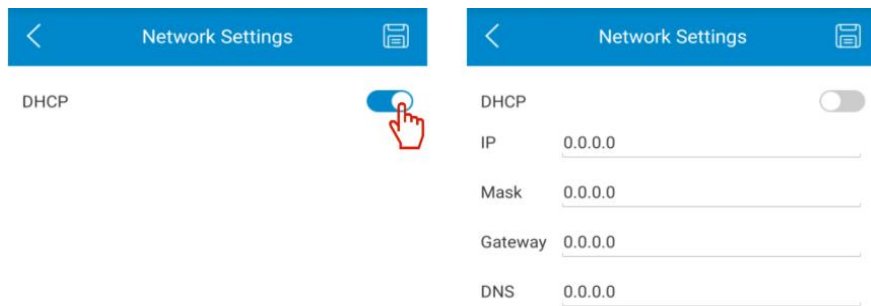
It also allows to change the settings and configure with customer server. In the very bottom of this page, there are also three buttons used to help users quickly complete the configuration. If connect the gateway to AWS IoT, please see [Appendix B Connect to AWS IoT](#) to get more details.

If configure with custom server, please follow these settings description as below:

Type	Parameter	Description
Broker setting	Host	The IP address or domain name of your server.
	Port	Server port.
	Client id	MQTT client id, each device connected to the server should have a different client id. The default id is device MAC address.
Topics	Subscribe	It has a default topic, can be changed
	Publish	It has a default topic, can be changed
General	Clean session	Default: Enable, range: Enable/Disable
	Qos	Quality of service. Default: 1, range: 0-2
	Keep Alive	Default: 60, range: 10-120
User Credentials	Username	If access to your server doesn't require a username and password, it can be blank.
	Password	
SSL/TLS	SSL/TLS	Enable: SSL mode. Disable: TCP mode
	Certificates	It supports three types of certificates: ➤ CA signed server certificate: no need to upload any certificate ➤ CA certificate file: need to upload CA Root certificate ➤ Self signed certificates: need to upload CA Root certificate, client key and client certificate

3.2.3 Network settings

To configure the IP allocation method for the gateway. It supports to use DHCP or statics IP, when disable DHCP, it will inactivate the statics IP.

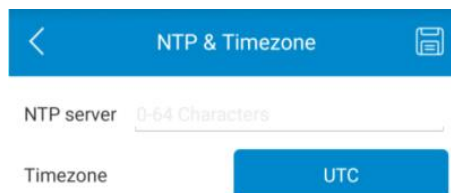


The image shows two screenshots of the 'Network Settings' interface. The left screenshot shows the 'DHCP' toggle switch being turned on, indicated by a red hand icon. The right screenshot shows the static IP configuration fields (IP, Mask, Gateway, DNS) which are inactive when DHCP is enabled.

3.2.4 NTP&Timezone

To configure the NTP server and timezone for the gateway.

The gateway has three default NTP servers, it allows users to configure one NTP server. If users configure it, the gateway will use the configured one, if not configure, the gateway will use default servers.



The image shows the 'NTP & Timezone' configuration interface. It features a text input field for the 'NTP server' with a placeholder '0-64 Characters' and a dropdown menu for the 'Timezone' currently set to 'UTC'.

3.2.5 Scanner filter

To configure scanning filter for the gateway, here we can set RSSI, one MAC address and one advertising name to filter Beacon data.

After the gateway is connected to MQTT broker, the filters can be set be MQTT commands.

<
Scanner Filter

Filter by RSSI (-127dBm~0dBm)

-127dBm

Filter by MAC Address

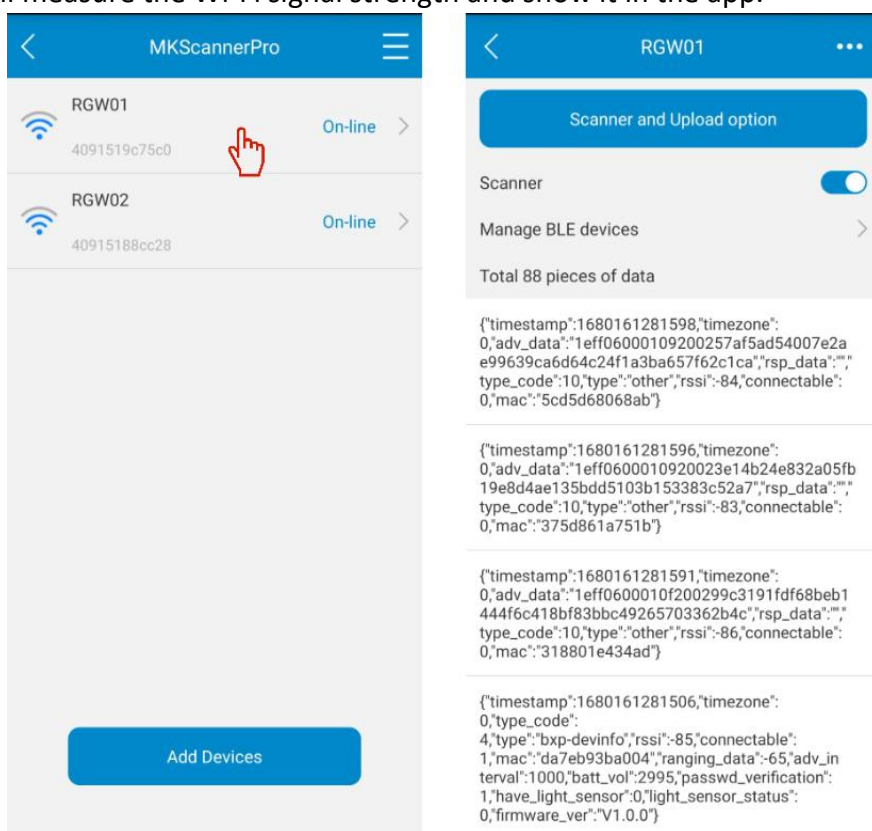
0~6 Bytes

Filter by ADV Name

0~20 Characters

3.3 Scan BLE device

When the gateway is successfully configured, it will automatically start scanning. At the same time, the gateway will measure the Wi-Fi signal strength and show it in the app.

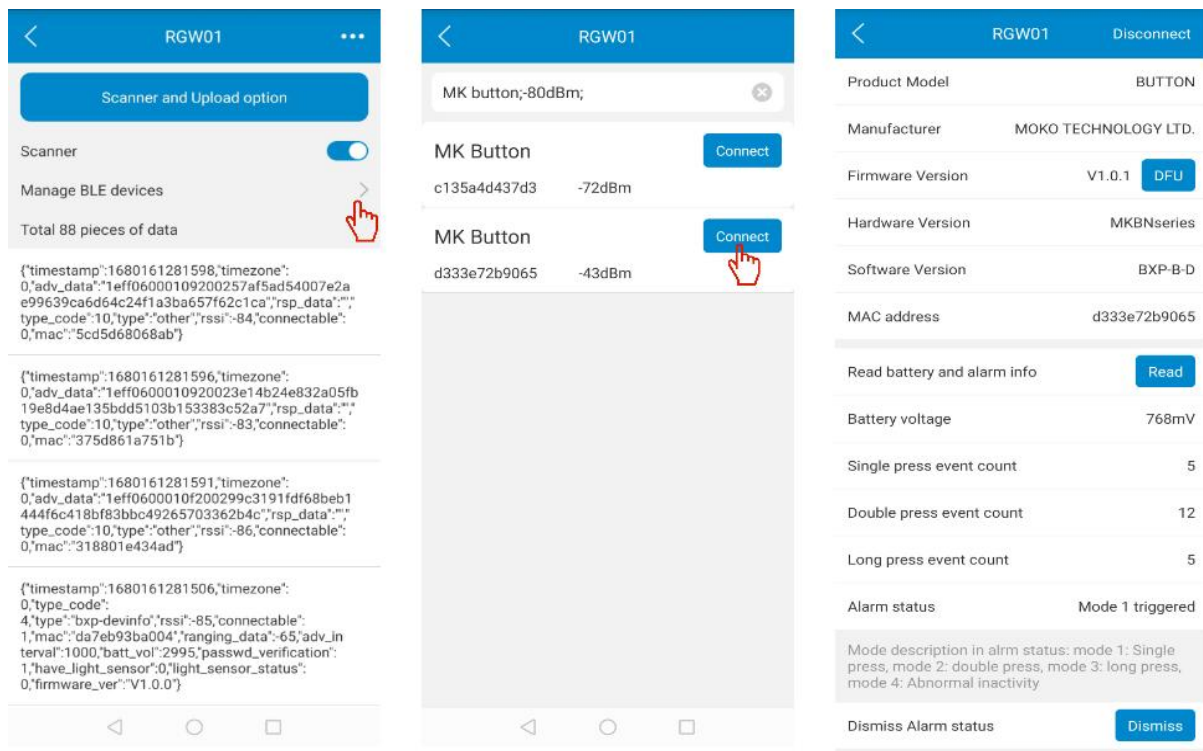


Icon	Wi-Fi signal strength	Wi-Fi RSSI
	OFFLINE	/
	POOR	<-65 dBm
	MEDIUM	-65~-50 dBm
	GOOD	>-50 dBm

3.4 Manage BLE devices

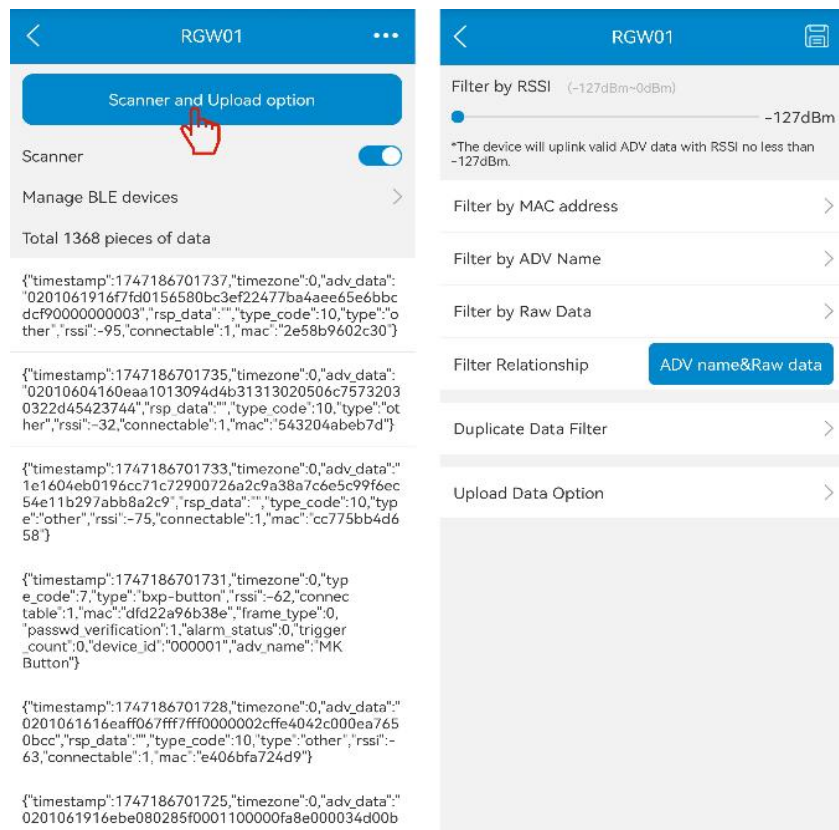
Click the manage BLE device button, it will jump to the next page where you can search and connect the nearby Beacon device.

We can research a MK Button device for connection. After the gateway is connected with the Beacon, we can get the product information, battery and alarm information, and dismiss the alarm status. The “disconnect” button in the top used to disconnect from the beacon.



3.5 Set scanner and upload option

To set the scanning filter and uploading data content for the gateway.



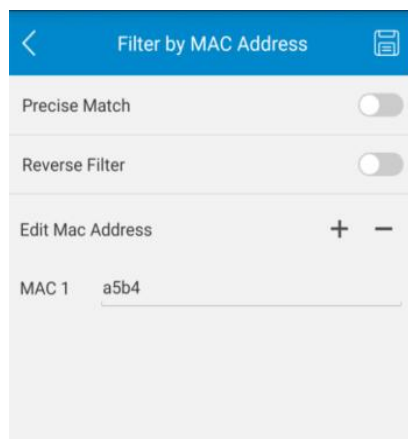
3.5.1 Filter By RSSI

The gateway will upload the beacon advertising data with RSSI no less than the setting value.

Parameter	Description
RSSI	Default: -127 dBm, range: -127~0 dBm.

3.5.2 Filter By MAC address

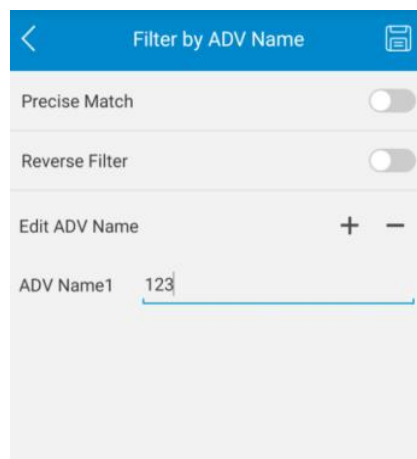
The gateway filters beacon data according to beacon MAC address, it supports up to 10 sets of MAC address at the same time.



Parameter	Description
Precise match	OFF: Upload the advertising data of the beacon whose MAC address contains the input expression. ON: Upload the advertising data of Beacon whose first N (N<=6) bytes of MAC is the same as the input expression.
Reverse filter	OFF: Upload advertising data of Beacon whose MAC address conforms the input expression. ON: Upload advertising data of Beacon whose MAC address doesn't conform the input expression.
MAC address	Click the “+” icon, it can add at most 10 sets of MAC address, the relationship of each MAC is “or”, and case insensitive. Click the “—” icon, it will delete the MAC address input box.

3.5.3 Filter By ADV Name

The gateway filters beacon data according to beacon advertising name, and it supports up to 10 set of advertising name at the same time.



Parameter	Description
Precise match	OFF: Upload the advertising data of the beacon whose adv name contains the input expression. ON: Upload the advertising data of Beacon whose first N (N<=20) bytes of adv name is the same as the input expression.
Reverse filter	OFF: Upload advertising data of Beacon whose adv name conforms the input expression. ON: Upload advertising data of Beacon whose adv name doesn't conform the input expression.
ADV name	Click the “+” icon, it can add at most 10 sets of advertising names, the relationship of each ADV name is “or”, and case insensitive. Click the “—” icon, it will delete the ADV name input box.

3.5.4 Filter By Raw Data

The gateway filters beacon data according to advertising data type.

The first 10 types are supported by MOKO beacon, the gateway will decode all MOKO beacon data. Other beacon data apart from the 10 types will be called “Other”, gateway will not decode the data, directly upload raw data to cloud.

Filter by Raw Data	
iBeacon	ON >
Eddystone-UUID	ON >
Eddystone-URL	ON >
Eddystone-TLM	ON >
BXP-Device info	☒
BXP - ACC	☒
BXP - T&H	☒
BXP - Button	ON >
BXP - Tag	ON >
PIR Presence	ON >
Other	ON >

3.5.4.1 iBeacon

To determine upload iBeacon data or not.

If the iBeacon switch is on, the iBeacon UUID, major and minor are empty, the gateway will upload all detected iBeacon data. If the iBeacon UUID, major and minor are filled with some value, the gateway will upload only the iBeacon data which conforms the value.

iBeacon	
iBeacon	☒
iBeacon UUID	
0~16 Bytes	
iBeacon Major	
Min	0 ~ Max 65535
iBeacon Minor	
Min	0 ~ Max 65535

Parameter	Description
Switch	On: upload, off: not upload
iBeacon UUID	0-16 bytes iBeacon UUID with Hex format
iBeacon major	From 0 to 65535, the Max value must be no less than the Min value.
iBeacon minor	From 0 to 65535, the Max value must be no less than the Min value.

3.5.4.2 Eddystone - UID

To determine upload Eddystone-UID data or not.

If the Eddystone-UID switch is on, the Namespace ID and Instance ID are empty, the gateway will upload all detected Eddystone - UID data. If the Namespace ID and Instance ID are filled with some value, the gateway will upload only the Eddystone - UID data which conforms the value.

Parameter	Description
Switch	On: upload, off: not upload
Namespace ID	0-10 bytes Hex data
Instance ID	0-6 bytes Hex data

3.5.4.3 Eddystone - URL

To determine upload Eddystone-URL data or not.

If the Eddystone - URL switch is on, the URL is empty, the gateway will upload all detected Eddystone - URL data. If the URL is filled with some value, the gateway will upload only the Eddystone - URL data which conforms the value.

Parameter	Description
Switch	On: upload, off: not upload
URL	0-37 characters, for example: www.mokosmart.com

3.5.4.4 Eddystone - TLM

To determine upload Eddystone-TLM data or not.

If the Eddystone - TLM switch is on, the TLM version is all, the gateway will upload all detected Eddystone - TLM data. If the TLM version is configured to 0 Or 1, the gateway will upload only the Eddystone - TLM data whose TLM version conforms the configuration.

Parameter	Description
Switch	On: upload, off: not upload
TLM version	Range: All/ version 0/ version 1 ➤ Null: All versions will be uploaded; ➤ Version 0: Unencrypted TLM; ➤ Version 1: Encrypted TLM

3.5.4.5 BXP- Device info

Parameter	Description
Switch	To determine upload BXP-device info data or not. On: upload, off: not upload

3.5.4.6 BXP- ACC

Parameter	Description
Switch	To determine upload BXP-ACC data or not. On: upload, off: not upload

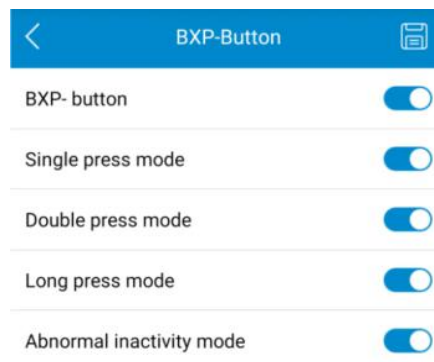
3.5.4.7 BXP- T&H

Parameter	Description
Switch	To determine upload BXP-T&H data or not. On: upload, off: not upload

3.5.4.8 BXP- Button

To determine upload BXP-Button data or not.

It supports filter the specified data by trigger modes. If all modes are on, the gateway will upload all detected BXP-button data. If the modes are off, the gateway will not upload the advertising data triggered by the modes.



3.5.4.9 BXP- Tag

To determine upload BXP-Tag data or not. It supports filter the specified data by Tag ID, allows to set up to 10 sets of Tag id at the same time.

Parameter	Description
Switch	On: upload, off: not upload
Precise match	OFF: Upload adverting data of Beacon whose tag id contains the input expression. ON: Upload adverting data of Beacon whose first N (N<=6) bytes of tag id is the same as the input expression.
Reverse filter	OFF: Upload advertising data of Beacon whose tag id conforms the input expression. ON: Upload advertising data of Beacon whose tag id doesn't conform the input expression.
Tag id	Click the “+” icon, it can add at most 10 sets of tag id, the relationship of each tag id is “or”, and case insensitive. Click the “—” icon, it will delete the tag id input box.

3.5.4.10 PIR Presence

To determine upload PIR Presence data or not.

It supports filter the specified data by sensor status and major, minor.

Parameter	Description
Switch	On: upload, off: not upload
Delay response status	Options: All/low delay/ medium delay/high delay All: All delay status advertising data will be uploaded
Door open/close response status	Options: All/close/open All: All door status advertising data will be uploaded
Sensor sensitivity	Options: All/low/medium/high All: All sensor sensitivity advertising data will be uploaded
Detection status	Options: All/no motion detected/motion detected All: All detection status advertising data will be uploaded
Major	From 0 to 65535, the Max value must be no less than the Min value.
Minor	From 0 to 65535, the Max value must be no less than the Min value.

3.5.4.11 Other

The Beacon data other than the above 10 types will be judged as “other”. The reporting format of other type is raw data, without decoder.

<

Other

Other

☒

Filter Condition

+

-

Condition A

Data Type

1-29

~

1-29

Byte

Raw Data Field

Condition B

Data Type

1-29

~

1-29

Byte

Raw Data Field

Condition C

Data Type

1-29

~

1-29

Byte

Raw Data Field

Filter Relationship

A | B | C

Parameter	Description
Switch	On: upload, off: not upload

Condition A	➤ Data type: 1 byte Bluetooth data type ➤ Data range: The start and end byte under the data type. It can be set to any two values from 1-29, the end value must be no less than the start value. ➤ Raw data field: Raw data value under the data type, and the data length should match the data range.
Condition B	The same as condition A
Condition C	The same as condition A
Filter relationship	The “AND/OR” logic setting for the conditions.

3.5.5 Filter Relationship

After the MAC filter, ADV name filter or raw data filter is set, it also needs to set the filter relationship, the relationship determines the processing logic.

If you set one or more filters, but relationship is set as “Null”, the filters will not take effect. The relationship should match the filters, otherwise there will be no data uploaded.

Parameter	Description
Relationship	Default: Null Range: Null/ Only MAC/ Only ADV name/ Only raw data/ ADV name & Raw data/ MAC & ADV name & Raw data/ ADV name Raw data/ ADV name& MAC.

3.5.6 Duplicate Data Filter

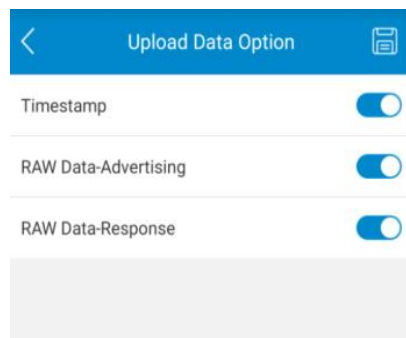
To reduce too many duplicate data uploaded to your server. In a filtering period, If the gateway scans a new data, it will report the data immediately, and throw the following data which are same as that one, finally report only one piece which is latest scanned in the period.

The screenshot shows a mobile application interface for configuring a 'Duplicate Data Filter'. At the top, there is a blue header bar containing a back arrow on the left, the text 'Duplicate Data Filter' in the center, and a save icon on the right. Below the header, the interface is divided into two sections. The first section, labeled 'Filter by', features a blue button with the text 'MAC'. The second section, labeled 'Filtering Period', contains a white input field with the number '10' and the unit 'Sec' to its right.

Parameter	Description
Filter by	Default: None, range: None/MAC/MAC + Data Type/MAC + Raw Data ➤ None: Duplicate data filter is disabled. ➤ MAC: Judge whether the data is duplicate according to the MAC address ➤ MAC+ Data Type: Judge whether the data is duplicate according to the MAC address and the data type. ➤ Mac+ Raw Data: Judge whether the data is duplicate according to the MAC address and the raw data.
Filtering Period	Only when the filter is enabled, the filtering period can be set. Default: 10, range: 1-86400 (Unit: second)

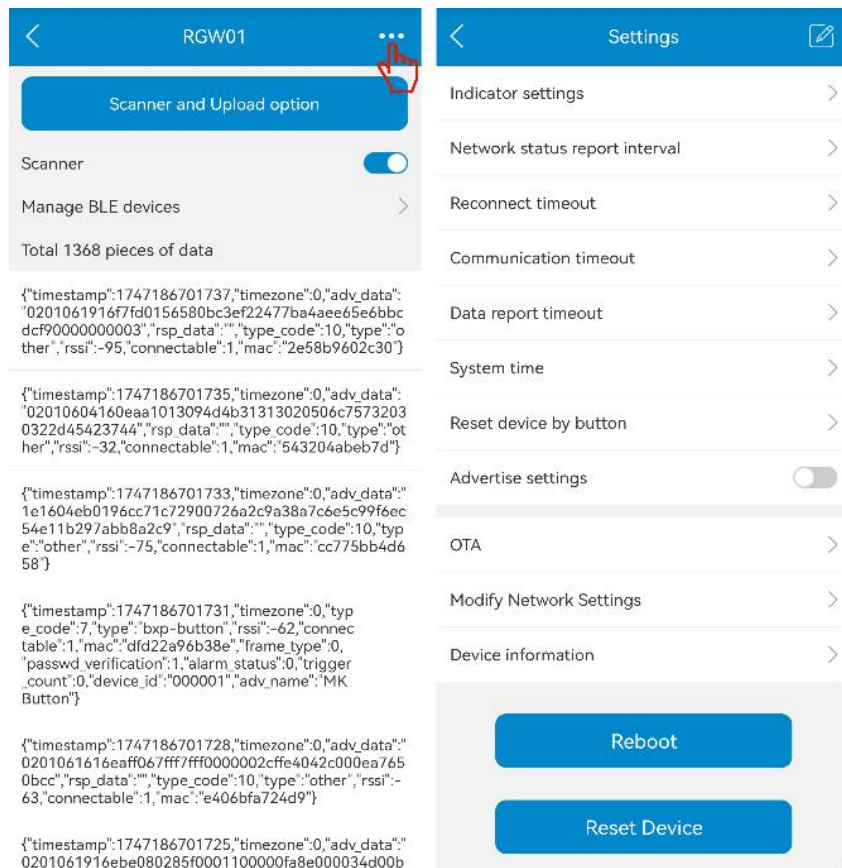
3.5.7 Upload Data Option

To determine the payload content uploaded to server. The Bluetooth data packet includes timestamp, MAC, RSSI, advertising raw data and response raw data. The timestamp and raw data are optional.



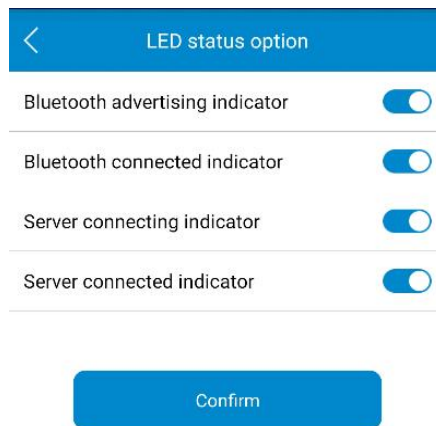
3.6 Gateway parameter settings

The gateway allows users to change its parameters. All parameters in the settings page can be modified.



3.6.1 LED Status Option

The LED indicator function in different device status can be configured.



Parameter	Description
Bluetooth advertising indicator	Default is enabled, when it is disabled, the LED will be OFF
Bluetooth connected indicator	Default is enabled, when it is disabled, the LED will be OFF
Server connecting indicator	Default is enabled, when it is disabled, the LED will be OFF
Server connected indicator	Default is enabled, when it is disabled, the LED will be OFF

3.6.2 Network Status Report Period

The gateway reports its network status to the server to notify the server that it is online. The report interval can be configured.

Parameter	Description
Network status report period	Default: 30, range: 0 or 10-86400 (unit: second) Value 0 means that the gateway will report the network status only once when it successfully connects to the server, will not report it later.

3.6.3 Reconnect Timeout

The gateway will automatically reboot once when it cannot connect to server in the configured timeout.

Parameter	Description
Connect timeout	Default: 3, range: 0-1440 (unit: minute) Value 0 means that the device will not reboot

3.6.4 Communication timeout

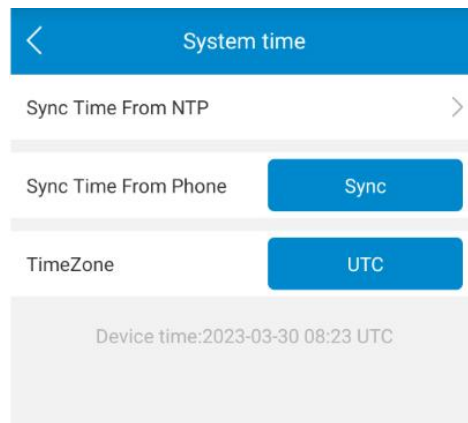
If the gateway doesn't get any downlink message from the cloud in the timeout, it will automatically disconnect from the beacon.

Parameter	Description
Communicate timeout	Default: 10, range: 0-60 (unit: minute) Value 0 means no automatic disconnection

3.6.5 System Time

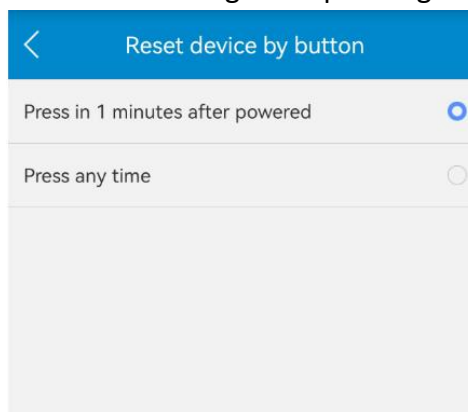
After the gateway connected with the server, it will synchronize time from the NTP server every 1 hour. If the NTP server is invalid, it also supports to synchronize time from user's phone.

The "Sync" button is used to require the UTC time from your phone, it also needs to select the TimeZone to obtain the local current time.



3.6.6 Reset by button

To choose the device reset mode. By default, the gateway can be reset by pressing button for 10 second in 1 minute after powered. We can change it to pressing button for 10 seconds any time.



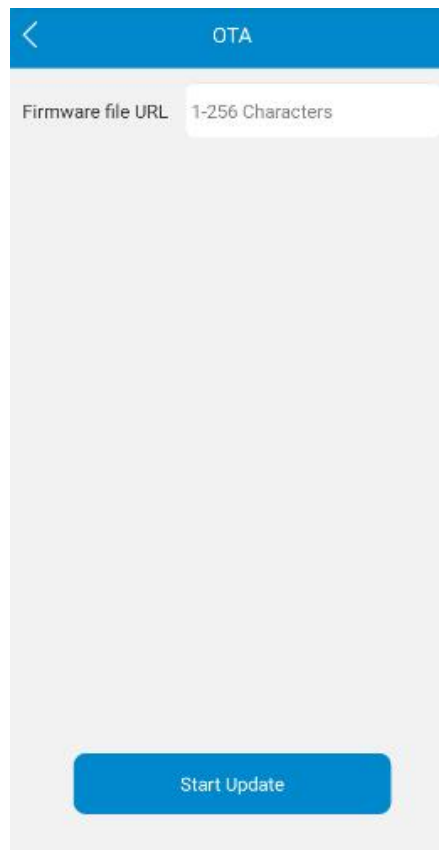
3.6.7 Advertising settings

To set the gateway advertise data. If click the switch to on, it requires to enter the advertising duration, the range is 1-10 minute. After that the gateway will advertise, the advertising name is the product model plus last 2 bytes of mac address.

3.7 OTA

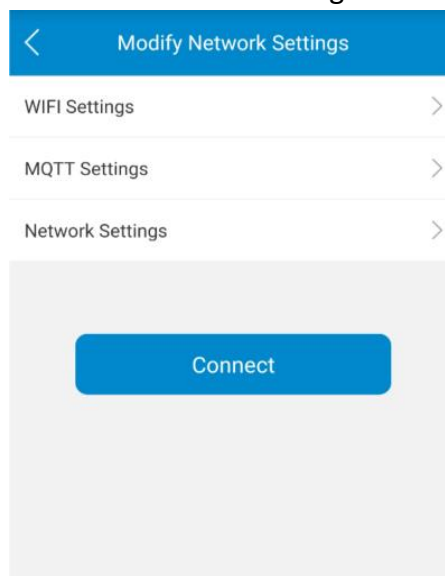
The gateway has an ability to update firmware over the air. When MOKO releases a new firmware, you can easily upgrade your gateway firmware by loading an upgrade Bin file with MOKO APP. The firmware file URL will be like: http://47.104.172.169:8080/updata_fold/MKGW_V1.0.4.bin

During upgrade process, LED will flash yellow, if upgrade succeed, LED turns solid yellow, if failed, LED turns solid red.



3.8 Modify Network settings

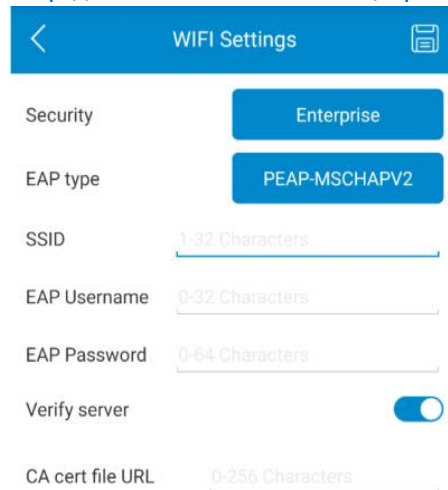
Change the WiFi, MQTT and network settings for the gateway, it allows users to change the settings independently. After the WIFI, MQTT or network settings are changed, click the “Connect” button, the gateway will reboot once and reconnect using the new settings.



3.8.1 Modify WiFi settings

To change the WIFI settings for the gateway. If we change security from personal to Enterprise, it requires to upload WIFI certificates, the certificates will be obtained from your HTTP server.

The CA cert file URL will be like: http://47.104.172.169:8080/updata_fold/wifi_ca.pem



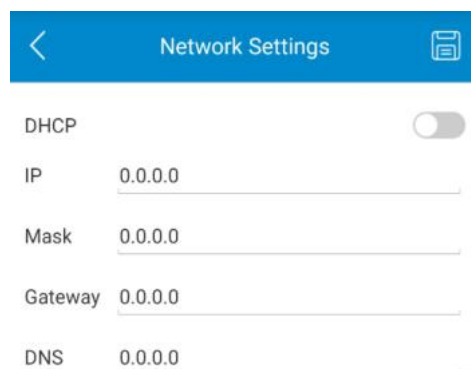
3.8.2 Modify MQTT settings

To change the MQTT settings for the gateway. If the new server requires SSL certificates, the certificates will be obtained from your HTTP server.

The certificates file URL will be like: http://47.104.172.169:8080/updata_fold/aws_ca.pem

3.8.3 Modify network settings

To change IP settings for the gateway, it supports DHCP and statics IP.



3.9 Device Information

You can get the device information in this page.

Device Information	
Device name	MK107-9F6E
Product Model	MK107
Manufacturer	MOKO TECHNOLOGY LTD.
Firmware Version	V1.0.4B
Software Version	V4.4.3
Hardware Version	V1.3.1.2
WIFI STA MAC	409151b99f6c
BT MAC	409151b99f6e

3.10 Reboot

The “Reboot” button on the screen is used to send a reboot command to the device. After that, the gateway will reboot once.

3.11 Reset Device

The “Reset Device” button on the screen is used to send a reset command to the device. After that, the device will restore to factory setting, and the indicator will flash blue and green once.

We can also press and hold the reset button for 10 seconds to reset it, please notice, with default settings, the reset will take effect only in 1minute after the gateway is powered up.

Appendix A: WPA2 Enterprise Security

Go to WIFI settings, select Enterprise for security, then select EAP type and enter the correct settings.

The image displays three side-by-side screenshots of the 'WIFI Settings' application interface, each showing a different EAP type selected for Enterprise security.

- Left Screenshot (PEAP-MSCHAPV2):** Shows the 'Security' dropdown set to 'Enterprise' and the 'EAP type' dropdown set to 'PEAP-MSCHAPV2'. Below, there are input fields for 'SSID' (1-32 Characters), 'EAP Username' (0-32 Characters), and 'EAP Password' (0-64 Characters). There is a 'Verify server' toggle switch (turned on) and a 'CA certificate' selection button (three dots).
- Middle Screenshot (TTLS-MSCHAPV2):** Shows the 'Security' dropdown set to 'Enterprise' and the 'EAP type' dropdown set to 'TTLS-MSCHAPV2'. Below, there are input fields for 'SSID' (1-32 Characters), 'EAP Username' (0-32 Characters), and 'EAP Password' (0-64 Characters). There is a 'Verify server' toggle switch (turned on) and a 'CA certificate' selection button (three dots).
- Right Screenshot (TLS):** Shows the 'Security' dropdown set to 'Enterprise' and the 'EAP type' dropdown set to 'TLS'. Below, there are input fields for 'SSID' (1-32 Characters) and 'Domain ID' (0-64 Characters). There are three certificate selection buttons: 'CA certificate', 'Client certificate', and 'Client key', each with a three-dot menu icon. A note at the bottom states: 'Please note the CA certificate is required, the client certificate and client key are optional.'

- PEAP-MSHCHAPV2: The user should enter the WIFI SSID, EAP username, EAP password. If verify server is enabled, it requires to upload the CA certificate from your phone.
- TTLS-MSHCHAPV2: The settings are the same as PEAP-MSHCHAPV2
- TTLS: The user should enter the WIFI SSID, domain ID, and upload certificates. The CA certificate is must-required, the client certificate and client key are optional.

Appendix B: Connect to AWS IoT

Go to Settings for APP page, firstly users should configure the APP connecting to AWS IoT. Then go to settings for device-> MQTT settings, configure the gateway connecting to AWS IoT.

The host is AWS host URL, port is 8883, turn on the SSL/TLS, select CA self signed certificates as the certificate type, then upload the CA certificate, client key and client cert file from your phone.

If using Android phone, the certificate files must be saved in the root directory of the local storage, otherwise the APP cannot obtain the files correctly.

Settings for APP

Broker Setting

Hosta1fhygr0xxahcm-ats.iot.us-west-2.ama
Port8883
Client IdMK_F2919D43

Topics

Subscribe0-128 Characters
Publish0-128 Characters

Note: The topic settings is only for connecting to Alibaba iot. Please leave it blank if connecting to a MQTT broker or AWS iot.

GeneralUser CredentialsSSL/TLS

SSL/TLS

CertificateSelf signed certificates

CA File/storage/emulated/0/.001/MOKO AWS file/zaws-root-ca.pem

Client Key/storage/emulated/0/.001/MOKO AWS file/zapp-private.pem.key

Client Cert File/storage/emulated/0/.001/MOKO AWS file/zapp-certificate.pem.crt

MQTT settings

Broker Setting

Host1ygr0xxahcm-ats.iot.us-west-2.amazon:
Port8883
Client Id4091519c75c0

Topics

Subscribe/MK110/4091519c75c0/receive
Publish/MK110/4091519c75c0/send

GeneralUser CredentialsSSL/TLS

SSL/TLS

CertificateSelf signed certificates

CA File/storage/emulated/0/.001/MOKO AWS file/zaws-root-ca.pem

Client Key/storage/emulated/0/.001/MOKO AWS file/zapp-private.pem.key

Client Cert File/storage/emulated/0/.001/MOKO AWS file/zapp-certificate.pem.crt

If using IOS phone, users need convert AWS certificates to required format, and import the certificates into your iOS phones through iTunes, then users can upload certificates from their phones.

Please note that only the APP settings require the converted certificate, while the device settings still use the CA root certificate, private key and client certificate original files.

Settings for APP

Broker Setting

Hosta1fhygr0xxahcm-ats.iot.us...
Port8883
Client IdMK_62316651

Topics

Subscribe0-128 Characters
Publish0-128 Characters

Note: The topic settings is only for connecting to Alibaba iot. Please leave it blank if connecting to a MQTT broker or AWS iot.

GeneralUser CredentialsSSL/TLS

SSL/TLS

CertificateSelf signed certificates

CA Fileaws-root-ca.der

P12 Cert Fileios-certificate file.p12

Clear All Configurations

MQTT settings

Broker Setting

Hosta1fhygr0xxahcm-ats.iot.us...
Port8883
Client Idc8f09ee8327c

Topics

Subscribe/MK110 plus 01/c8f09ee83...
Publish/MK110 plus 01/c8f09ee83...

GeneralCredentialsSSL/TLSLWT

SSL/TLS

CertificateSelf signed certificates

CA Filezaws-root-ca.pem

Client Keyzdevice-private.pem.key

Client Cert Filezdevice-certificate.pem.crt

Clear All Configurations

Appendix C: Configuration APP SDK

MOKO provides the configuration APP code, users can download the source code from the below links, and develop your own APP based on the MOKO SDK.

MK107/MKGW-mini 01/MK110Plus 01:

Android: https://github.com/MKScannerPro/Remote_Gateway_32_Android.git

IOS: https://github.com/MKScannerPro/Remote_Gateway_32_iOS.git

MK107Pro-32D:

Android: https://github.com/MKScannerPro/MK107Pro-32D_Android.git

IOS: https://github.com/MKScannerPro/MK107Pro-32D_IOS.git

MKGW-mini 02-32D:

Android: https://github.com/MKScannerPro/MKGW-mini-02-32D_Android.git

IOS: https://github.com/MKScannerPro/MKGW-mini-02-32D_IOS.git

MKGW-mini 03-20D:

Android: https://github.com/MKScannerPro/MKGW-mini03-20D_Android.git

IOS: https://github.com/MKScannerPro/MKGW-mini03-20D_IOS.git

Revision History

Revision	Description	Editor	Date
V1.0	Initial version	Weiguifen	2023.4.20
V1.1	Improve document	Weiguifen	2023.9.11
V1.2	Added applied product models and configuration APP SDK	Weiguifen	2024.3.14
V1.3	Add applied gateway product model, and differences of gateway models.	Weiguifen	2025.5.12
V1.4	Add APP SDK for different gateways	Weiguifen	2025.5.22

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